AIM:

To write a Python program to solve the Vacuum Cleaner problem using the Artificial Intelligence technique.

ALGORITHM:

1. Start the program.
2. Initialize two rooms A and B as dirty.
3. Place the vacuum cleaner in room A.
4. Check the current room condition.
5. If the room is dirty, clean it.
6. If the room is clean, move to the next room.
7. Repeat the process until all rooms are clean.
8. Display the final status of the rooms.
9. Stop the program.

PROGRAM :

pgrm6.py - C:/Users/nivet/AppData/Local/Programs/Python/Python311/pgrm6.py (3.11.0)

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```
# Vacuum Cleaner Problem

def vacuum_cleaner():
    room = {
        'A': 'Dirty',
        'B': 'Dirty'
    }

    location = 'A'

    while 'Dirty' in room.values():

        print("Vacuum is in Room", location)

        if room[location] == 'Dirty':
            print("Cleaning Room", location)
            room[location] = 'Clean'

        elif location == 'A':
            location = 'B'
            print("Moving to Room B")

        elif location == 'B':
            location = 'A'
            print("Moving to Room A")

    print("\nAll rooms are clean!")
    print(room)


vacuum_cleaner()
```

OUTPUT :

```
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>
=== RESTART: C:/Users/nivet/AppData/Local/Programs/Python/Python311/pgrm6.py ===
Vacuum is in Room A
Cleaning Room A
Vacuum is in Room A
Moving to Room B
Vacuum is in Room B
Cleaning Room B

All rooms are clean!
{'A': 'Clean', 'B': 'Clean'}
>>
```

RESULT :

Thus, the Python program to solve the Vacuum Cleaner problem was executed successfully, and all rooms were cleaned.